D2 Debugging tools

breakpoint(), pdb, logging levels, Spark UI and explain().

D2-L1 breakpoint() and pdb, stop and look

Let the crash open the debugger, print the failing frame's inputs, then walk up until the values go wrong.

```
python -m pdb -c continue run_qa.py
...
ZeroDivisionError: float division by zero
Uncaught exception. Entering post mortem debugging
> /repo/run_qa.py(2)weighted_pct()
-> ratio = reach / base
(Pdb) p reach, base
(5.0, 0.0)
```

Open pdb where a script crashed	python -m pdb -c continue script.py. You land in the frame that raised.
pdb: print, stack, up, next, step, continue	p, w, u, n, s, c.
PYTHONBREAKPOINT=0	Makes breakpoint() a no-op. A safety net for jobs, not a reason to commit breakpoints.
Why a breakpoint in a Spark UDF never fires	UDFs run on executors. The debugger stops only the driver. Test UDF logic as plain Python.

D2-L2 Logging levels that help at 3 a.m.

One named logger per module, levels that match severity, identifiers in every line, and a raise after every CRITICAL.

```
import logging

logger = logging.getLogger(__name__)
```

Logging levels and numbers	DEBUG 10, INFO 20, WARNING 30, ERROR 40, CRITICAL 50. Default effective level: WARNING.
basicConfig called twice	The second call does nothing, unless force=True.
logger.exception	Logs at ERROR with the traceback. Call it inside an except block.
Does logger.critical stop the pipeline?	No. Log, then raise.

D2-L3 explain(), the Spark UI and the query profile

Read the plan bottom up, look for pushed filters, shuffles, join types and Python UDFs, then confirm rows per operator in the Spark UI or query profile.

Read a physical plan in which direction?	Bottom up: scan at the bottom, result at the top.
PartitionFilters and PushedFilters	Filters applied while reading. Empty PushedFilters on a filtered query means no pushdown.
Plan operators that signal trouble	CartesianProduct or BroadcastNestedLoopJoin (cross join), BatchEvalPython (Python UDF), many Exchange (shuffles).

Spark UI on serverless	Not available. Use the query profile: time spent, rows and memory per operator.